

FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING
Department of Computer Engineering
Academic Year 2025-26

RUBRICS

Class : B.E. Computer Engineering
Subject Code : CSL702

Subject Name : BDA Lab Semester : VII

<u>Practical No:</u>	9
<u>Title:</u>	Write queries to sort and aggregate the data in a table using HiveQL
<u>Date of Performance:</u>	17/10/2025
<u>Roll No:</u>	9913
<u>Name of the Student:</u>	Mark Xavier Lopes

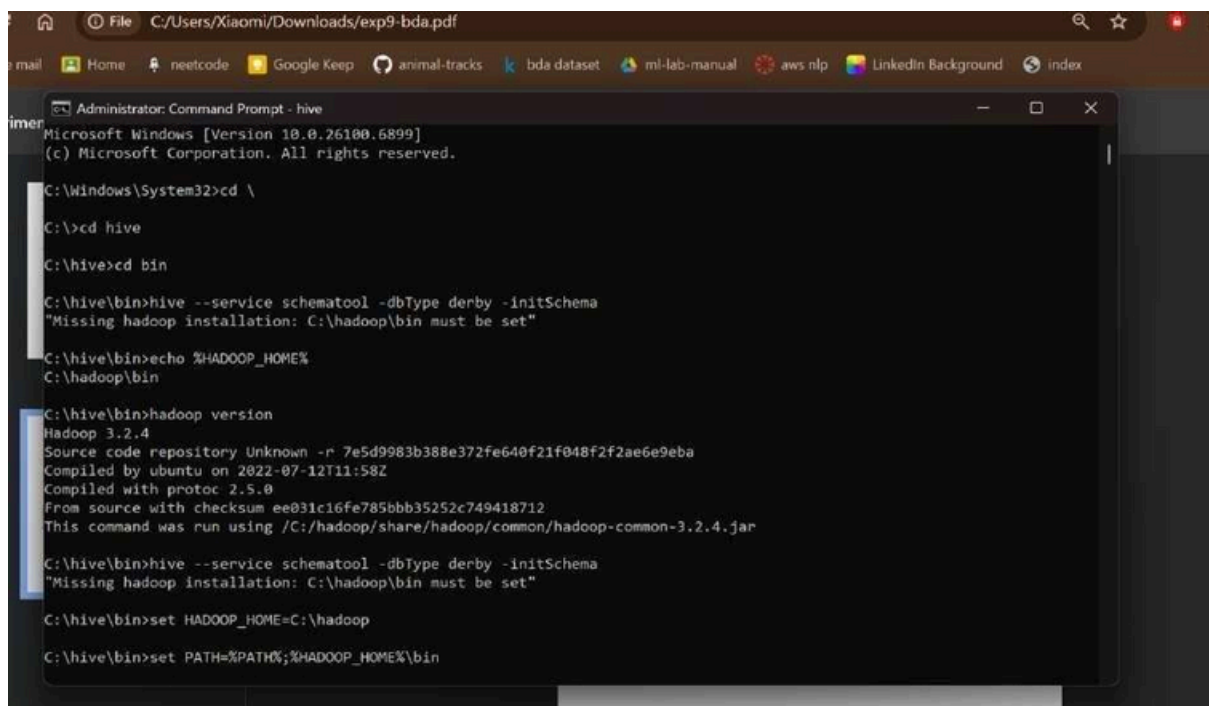
Evaluation:

<u>Performance Indicator</u>	<u>Below average</u>	<u>Average</u>	<u>Good</u>	<u>Excellent</u>	<u>Marks</u>
<u>On Time Submission (2)</u>	<u>Not submitted (0)</u>	<u>Submitted after deadline (1)</u>	<u>Early or on time submission (2)</u>	<u>---</u>	
<u>Test cases and output (4)</u>	<u>Incorrect output (1)</u>	<u>The expected output is verified only a for few test cases (2)</u>	<u>The expected output is Verified for all test cases but is not presentable (3)</u>	<u>Expected output is obtained for all test cases. Presentable and easy to follow (4)</u>	
<u>Coding efficiency (2)</u>	<u>The code is not structured at all (0)</u>	<u>The code is structured but not efficient (1)</u>	<u>The code is Structured and efficient. (2)</u>	<u>=</u>	
<u>Knowledge (2)</u>	<u>Basic concepts not clear (0)</u>	<u>Understood the basic concepts (1)</u>	<u>Could explain the concept with suitable example (1.5)</u>	<u>Could relate the theory with real world applicaion (2)</u>	
<u>Total</u>					

Signature of the Teacher:

Aim: Write queries to sort and aggregate the data in a table using HiveQL Objective: The objective of this experiment is to demonstrate how to write queries to sort and aggregate data stored in a Hive table using HiveQL Tools and Technologies:

- Hive: A data warehouse infrastructure built on top of Hadoop for providing data summarization, query, and analysis.
- HiveQL: Hive Query Language, which is similar to SQL and used for querying and managing data in Hive.



```
Administrator: Command Prompt - hive
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd \

C:\>cd hive

C:\hive>cd bin

C:\hive\bin>hive --service schematool -dbType derby -initSchema
"Missing hadoop installation: C:\hadoop\bin must be set"

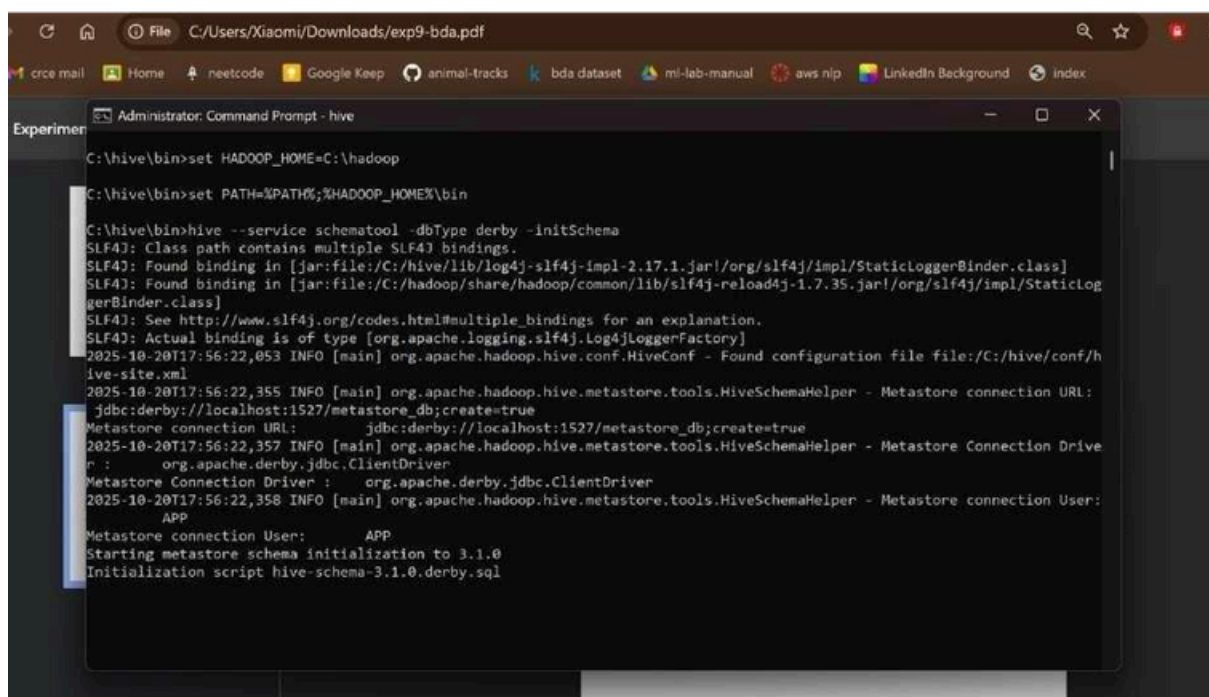
C:\hive\bin>echo %HADOOP_HOME%
C:\hadoop\bin

C:\hive\bin>hadoop version
Hadoop 3.2.4
Source code repository Unknown -r 7e5d9983b388e372fe640f21f048f2f2ae6e9eba
Compiled by ubuntu on 2022-07-12T11:58Z
Compiled with protoc 2.5.0
From source with checksum ee031c16fe785bbb35252c749418712
This command was run using /C:/hadoop/share/hadoop/common/hadoop-common-3.2.4.jar

C:\hive\bin>hive --service schematool -dbType derby -initSchema
"Missing hadoop installation: C:\hadoop\bin must be set"

C:\hive\bin>set HADOOP_HOME=C:\hadoop

C:\hive\bin>set PATH=%PATH%;%HADOOP_HOME%\bin
```



```
Administrator: Command Prompt - hive

C:\hive\bin>set HADOOP_HOME=C:\hadoop

C:\hive\bin>set PATH=%PATH%;%HADOOP_HOME%\bin

C:\hive\bin>hive --service schematool -dbType derby -initSchema
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/C:/hive/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/C:/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.35.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
2025-10-20T17:56:22,053 INFO [main] org.apache.hadoop.hive.conf.HiveConf - Found configuration file file:/C:/hive/conf/hive-site.xml
2025-10-20T17:56:22,355 INFO [main] org.apache.hadoop.hive.metastore.tools.HiveSchemaHelper - Metastore connection URL: jdbc:derby://localhost:1527/metastore_db;create=true
Metastore connection URL: jdbc:derby://localhost:1527/metastore_db;create=true
2025-10-20T17:56:22,357 INFO [main] org.apache.hadoop.hive.metastore.tools.HiveSchemaHelper - Metastore Connection Driver: org.apache.derby.jdbc.ClientDriver
Metastore Connection Driver: org.apache.derby.jdbc.ClientDriver
2025-10-20T17:56:22,358 INFO [main] org.apache.hadoop.hive.metastore.tools.HiveSchemaHelper - Metastore connection User: APP
Metastore connection User: APP
Starting metastore schema initialization to 3.1.0
Initialization script hive-schema-3.1.0.derby.sql
```

```
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Xiaomi>set HADOOP_HOME=C:\hadoop

C:\Users\Xiaomi>set PATH=%PATH%;%HADOOP_HOME%\bin;%HADOOP_HOME%\sbin

C:\Users\Xiaomi>cd C:\hive\bin

C:\hive\bin>hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/C:/hive/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/C:/hadoop/share/hadoop/common/lib/slf4j-reload4j-1.7.35.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
2025-10-20T19:25:03,303 INFO [main] org.apache.hadoop.hive.conf.HiveConf - Found configuration file file:/C:/hive/conf/hive-site.xml
Hive Session ID = f659d96b-da36-4ec5-9c0f-ba14be2c668a

Logging initialized using configuration in jar:file:/C:/hive/lib/hive-common-3.1.3.jar!/hive-log4j2.properties Async: true
2025-10-20T19:25:04,037 INFO [main] org.apache.hadoop.hive ql.session.SessionState - Created HDFS directory: file:/C:/hive/tmp/Xiaomi/f659d96b-da36-4ec5-9c0f-ba14be2c668a
2025-10-20T19:25:04,041 INFO [main] org.apache.hadoop.hive ql.session.SessionState - Created local directory: C:/hive/tmp/f659d96b-da36-4ec5-9c0f-ba14be2c668a
2025-10-20T19:25:04,042 INFO [main] org.apache.hadoop.hive ql.session.SessionState - Created HDFS directory: file:/C:/hive/tmp/Xiaomi/f659d96b-da36-4ec5-9c0f-ba14be2c668a/_tmp_space.db
2025-10-20T19:25:04,052 INFO [main] org.apache.hadoop.hive.conf.HiveConf - Using the default value passed in for log id: f659d96b-da36-4ec5-9c0f-ba14be2c668a
```

```
Logging initialized using configuration in jar:file:/C:/hive/lib/hive-common-3.1.3.jar!/hive-log4j2.properties Async: true
2025-10-20T19:25:04,037 INFO [main] org.apache.hadoop.hive ql.session.SessionState - Created HDFS directory: file:/C:/hive/tmp/Xiaomi/f659d96b-da36-4ec5-9c0f-ba14be2c668a
2025-10-20T19:25:04,041 INFO [main] org.apache.hadoop.hive ql.session.SessionState - Created local directory: C:/hive/tmp/f659d96b-da36-4ec5-9c0f-ba14be2c668a
2025-10-20T19:25:04,042 INFO [main] org.apache.hadoop.hive ql.session.SessionState - Created HDFS directory: file:/C:/hive/tmp/Xiaomi/f659d96b-da36-4ec5-9c0f-ba14be2c668a/_tmp_space.db
2025-10-20T19:25:04,052 INFO [main] org.apache.hadoop.hive.conf.HiveConf - Using the default value passed in for log id: f659d96b-da36-4ec5-9c0f-ba14be2c668a
2025-10-20T19:25:04,052 INFO [main] org.apache.hadoop.hive ql.session.SessionState - Updating thread name to f659d96b-da36-4ec5-9c0f-ba14be2c668a main
hive> |
```

```
2025-10-20T19:25:04,052 INFO [main] org.apache.hadoop.hive ql.session.SessionState - Updating thread name to f659d96b-da36-4ec5-9c0f-ba14be2c668a main
hive> Display all 634 possibilities? (y or n)
!
$KEY$
$key$
%
)
+
.
<
<>
>
ACTIVATE
ALLOC_FRACTION
ARRAY
BIGINT
BUCKET
CAST
CLUSTERED
COMMENT
CONSTRAINT
DATE
DEFINED
DESCRIBE
DISTINCT
DOUBLE
EXCEPT
EXTERNAL
FLOAT
!=
$VALUE$
$sum0
&
);
/
<=
=
>=
ABORT
ADD
ALTER
AS
BINARY
BUCKETS
CHECK
COLLECTION
COMPACT
CREATE
DATETIME
DELIMITED
DIRECTORY
DISTRIBUTE
DROP
EXPLAIN
FALSE
FOREIGN
$LEMS$
$lem$
$value$
(
*
-
/
:
<=>
==
ABORT
ALL
AND
ASC
BOOLEAN
BY
CLUSTER
COLUMNS
COMPACTIONS
DATA
DEFAULT
DESC
DISABLE
DO
ENABLE
EXTENDED
FIELDS
FORMAT
```

```
//> create database hivesql;

OK

No rows affected (1.183 seconds)
```

```

    /> create table hivesql.transactions(acct int, amount decimal(18,2), txn_date
date);
OK
No rows affected (2.096 seconds)

```

```

insert into hivesql.transactions(acct,amount,txn_date)
values
(101,10.01, DATE'2021-01-01'),
(101,102.01, DATE'2021-01-01'),
(102,93., DATE'2021-01-01'),
(103,913.1, DATE'2021-01-02'),
(102,913.1, DATE'2021-01-02'),
(101,900.56, DATE'2021-01-03');

```

```

    //> select * from hivesql.transactions;
OK
+-----+-----+-----+
| transactions.acct | transactions.amount | transactions.txn_date |
+-----+-----+-----+
| 101               | 10.01               | 2021-01-01           |
| 101               | 102.01              | 2021-01-01           |
| 102               | 93.00               | 2021-01-01           |
| 103               | 913.10              | 2021-01-02           |
| 102               | 913.10              | 2021-01-02           |
| 101               | 900.56              | 2021-01-03           |
+-----+-----+-----+
6 rows selected (0.38 seconds)

```

```

Logging initialized using configuration in jar:file:/C:/hive/lib/hive-common-3.1.3.jar!/hive-log4j2.properties Async: true
2025-10-20T19:46:03,945 INFO [main] org.apache.hadoop.hive.ql.session.SessionState - Created HDFS directory: file:/C:/hive/tmp/Xiaomi/f3de014c-421c-458b-9c4d-234598e6bd2d
2025-10-20T19:46:03,947 INFO [main] org.apache.hadoop.hive.ql.session.SessionState - Created local directory: C:/hive/tmp/f3de014c-421c-458b-9c4d-234598e6bd2d
2025-10-20T19:46:03,948 INFO [main] org.apache.hadoop.hive.ql.session.SessionState - Created HDFS directory: file:/C:/hive/tmp/Xiaomi/f3de014c-421c-458b-9c4d-234598e6bd2d/_tmp_space.db
2025-10-20T19:46:03,958 INFO [main] org.apache.hadoop.hive.conf.HiveConf - Using the default value passed in for log id: f3de014c-421c-458b-9c4d-234598e6bd2d
2025-10-20T19:46:03,958 INFO [main] org.apache.hadoop.hive.ql.session.SessionState - Updating thread name to f3de014c-421c-458b-9c4d-234598e6bd2d main
hive> SELECT TXN.*, LEAD(amount,1) OVER (PARTITION BY acct ORDER BY txn_date) AS amt_sub
> FROM hivesql.transactions as TXN;

```


txn.acct	txn.amount	txn.txn_date	amt_pre
101	102.01	2021-01-01	NULL
101	10.01	2021-01-01	102.01
101	900.56	2021-01-03	10.01
102	93.00	2021-01-01	NULL
102	913.10	2021-01-02	93.00
103	913.10	2021-01-02	NULL

```

2025-10-20T19:51:22,997 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.
e default value passed in for log id: f3de014c-421c-458b-9c4d-234598e6bd2d
2025-10-20T19:51:22,998 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.
- Resetting thread name to main
hive> SE
> SELECT COUNT(*) AS total_records,
> COUNT(distinct acct) as total_accounts
> FROM hiveql.transactions as TXN

```

total_records	total_accounts
6	3

```

2025-10-20T19:55:24,786 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.co
e default value passed in for log id: f3de014c-421c-458b-9c4d-234598e6bd2d
2025-10-20T19:55:24,786 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.ql
- Resetting thread name to main
hive> SELECT acct, SUM(amount) as total_amount
> FROM hivesql.transactions as TXN
> GROUP BY acct;

```

acct	total_amount
101	1012.58
102	1006.10
103	913.10

```

2025-10-20T19:56:56,377 INFO [main] org.apache.hadoop.hive.ql.session.SessionState - Updating thread name to f3de014c-421c-458b-9c4d-234598e6bd2d main
FAILED: HiveException java.lang.RuntimeException: Unable to instantiate org.apache.hadoop.hive.ql.metadata.SessionStoreClient
2025-10-20T19:59:51,221 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.conf.HiveConf - e default value passed in for log id: f3de014c-421c-458b-9c4d-234598e6bd2d
2025-10-20T19:59:51,223 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.ql.session.SessionState - Resetting thread name to main
hive> SELECT TXN.*,
> RANK() OVER (ORDER BY txn_date) as row_rank
> FROM hivesql.transactions as TXN;

```

txn.acct	txn.amount	txn.txn_date	row_rank
102	93.00	2021-01-01	1
101	102.01	2021-01-01	1
101	10.01	2021-01-01	1
102	913.10	2021-01-02	4
103	913.10	2021-01-02	4
101	900.56	2021-01-03	6

```

taStoreClient
2025-10-20T20:04:30,882 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.conf.HiveConf - e default value passed in for log id: f3de014c-421c-458b-9c4d-234598e6bd2d
2025-10-20T20:04:30,882 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.ql.session.SessionState - Resetting thread name to main
hive> select * from hivesql.transactions ORDER BY acct ASC, txn_date;

```

exp8-ml-9881.pdf	18-10-2025 23:22	Chrome PDF Docum...	1,290 KB
lab8-ml.pdf	18-10-2025 23:19	Chrome PDF Docum...	1,137 KB
exp7-ml-9881-compressed.pdf	18-10-2025 23:18	Chrome PDF Docum...	5,765 KB

transactions.acct	transactions.amount	transactions.txn_date
101	102.01	2021-01-01
101	10.01	2021-01-01
101	900.56	2021-01-03
102	93.00	2021-01-01
102	913.10	2021-01-02
103	913.10	2021-01-02

```
f3de014c-421c-458b-9c4d-234598e6bd2d
025-10-20T20:01:34,951 INFO [main] org.apache.hadoop.hive.ql.session.SessionState - Updating thread name to f3de0
c-458b-9c4d-234598e6bd2d main
FAILED: HiveException java.lang.RuntimeException: Unable to instantiate org.apache.hadoop.hive.ql.metadata.Session
aStoreClient
025-10-20T20:04:30,882 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.conf.HiveConf - Us
default value passed in for log id: f3de014c-421c-458b-9c4d-234598e6bd2d
025-10-20T20:04:30,882 INFO [f3de014c-421c-458b-9c4d-234598e6bd2d main] org.apache.hadoop.hive.ql.session.Session
Resetting thread name to main
hive> select * from hivesql.transactions SORT BY acct ASC, txn_date;|
| 103 | 913.10 | 2021-01-02 |
+-----+-----+-----+
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Config
```

transactions.acct	transactions.amount	transactions.txn_date
101	102.01	2021-01-01
101	10.01	2021-01-01
101	900.56	2021-01-03
102	93.00	2021-01-01
102	913.10	2021-01-02
103	913.10	2021-01-02

Observation:

During the experiment, different HiveQL queries were executed to perform sorting and aggregation operations on the data. Sorting queries successfully arranged records in ascending and descending order, while aggregation functions like COUNT, SUM, AVG, MIN, and MAX provided summarized insights. Combining sorting with aggregation helped in ranking and grouping data efficiently. The execution time varied slightly depending on the dataset size and query complexity.

Conclusion:

This experiment demonstrated how HiveQL can be used to sort and aggregate large datasets efficiently. It showed that Hive simplifies data summarization and analysis in a distributed environment, making it easier to extract meaningful insights from big data stored in Hadoop.